NARAYANAN KRISHNAMURTHY

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SUMMARY

Data science and analytics leader with 10+ years of experience using SQL, Python, R, machine learning, and business int elligence to improve business processes & operational decisions. Drive data driven insights in regulated, data-rich environ ments. Delivered measurable improvement in self-service analytics adoption, document retrieval intelligence, and revenue protection strategies. Skilled in collaborating cross-functionally to translate complex data into actionable results.

WORK EXPERIENCE

Internal Revenue Service (IRS) **Mar 2021 - Present**
Statistician *Plantation, FL*

Built SQL, R, Python, Tableau, and emerging ArcGIS solutions that improved workload visibility, semantic grouping, text analytics, RAG experimentation, and privacy-safe AI/ML development across LB&I practice areas.

- Partner with leaders, stakeholders, and developers to translate strategic priorities into current-state briefings, data requirements, analytical workstreams, and decision-support products spanning workload planning, geospatial analytics, AI/ML, and SEC text analytics.
- Engineered reproducible SQL, Python, and R pipelines across approximately 11 million corporate and partnership returns; expanded TY2024 queries covering approximately 8.5 million tax form records and created SQL Server assets for Tableau and ArcGIS products.
- Advanced automation and operational resilience by adding checkpoint-and-restart processing to long-running geocoding and developing ArcPy nearest-distance and buffer analyses to align available work and field resources across locations.
- Strengthened privacy-safe AI/ML evaluation and governance through containerized environments, metric selection, cross-language solution testing, KPI reviews, and early identification of source-data limitations.
- Validated analytical consistency by testing R and Python Automated Candidate Review implementations, investigating bag-of-words quantile differences, documenting functional parity, and peer-reviewing metric and KPI logic.
- Modernized semantic clustering by evaluating SBERT sentence embeddings, including MiniLM-L6-v2 and e5-base-v2, as supportable alternatives to a legacy ELMo/TensorFlow workflow.
- Expanded analytics and AI adoption through solution demonstrations, onboarding support, peer reviews, and cross-agency forums serving 3 communities of practice and 50-150 analysts weekly.

Frontier Communications **Jul 2020 - Mar 2021**
Data Science Analyst *Coconut Creek, FL*

Analyzed high-volume customer-care call data across a 10-month period using Teradata SQL, R, and pandas to identify frequent callers, interaction patterns, and service drivers for operational reporting.

- Improved customer record-linkage accuracy by approximately 2% by matching calls to accounts using caller ID, partial addresses, and multiple phone numbers, enabling more reliable customer-level analysis.
- Applied scikit-learn clustering and hypothesis testing to segment caller groups and validate differences, supporting more targeted service strategies and performance measurement.

Automated daily forecast reporting with C#/.NET, Microsoft Task Scheduler, and Excel, improving consistency and reducing manual reporting effort.

Fabmatics USA (Client: Magic Leap) **Feb 2019 - Dec 2019**
Software Engineer *Plantation, FL*

Supported Magic Leap's AR glasses prototyping environment by analyzing gRPC, PLC, and Linux system logs, coordinating issue translation between client and offshore German engineers, and helping sustain day-to-day operations across two 800-meter multistage processing lines.

- Analyzed gRPC, PLC, and Linux system logs to accelerate issue identification in a multi-vendor manufacturing environment.

Cognizant Technology Solutions (Client: Florida Power & Light) **Jun 2018 - Feb 2019**
Data Science Analyst *Jupiter, FL*

Supported FPL's Miami revenue protection team and data leads by applying R, Python, SPSS, and Postgres SQL to smart meter data, helping strengthen anomaly analysis and workflow-ready analytical outputs across a high-volume utility environment.

- Enhanced revenue analytics and protection analysis across ~2 million customer accounts by applying predictive modeling and year-over-year time series analysis to smart meter data, improving detection of abnormal usage behavior impacting revenue.
- Improved workflow readiness by building repeatable analysis on Postgres smart meter data, leading to a more usable and structured process for anomaly evaluation.
- Embedded change point analysis into smart meter evaluation by identifying where usage patterns materially shifted over time, leading to stronger anomaly detection within the existing workflow.

ORAU (Client: DOE National Energy Technology Laboratory) **Feb 2016 - Jun 2018**
Data Science Analyst *Pittsburgh, PA*
Helped Principal Investigators and research collaborators by applying machine learning and statistical analysis to complex materials datasets, guiding alloy evaluation and research exploration.

- Enabled more structured evaluation of 9Cr steel test data by building R- and Python-based analytical tools across DOE and Japanese datasets, leading to stronger exploration of alloy properties and experimental results.
- Strengthened alloy-property analysis by applying multivariate regression, PCA, variable importance analysis, and random forest-based modeling to existing test data, leading to better assessment of composition-property relationships.
- Performed clustering analysis on alloy to identify specific segments to better model properties within segments
- Aligned DOE and Case Western collaborators through research discussions and a white paper, improving interpretation of complex analytical findings.

IMPACT with ML & Embedded Projects

- SEC MD&A intelligence: Built a working RAG prototype over 2024–2025 public-company filings, fusing structured financial tables with narrative text, cross-walking records to NAICS industries, and using structure-aware chunking with SQLite structured + vector retrieval and Redis-ready interfaces.
- Synthetic-data evaluation: Built DGX containers and evaluated quality/privacy metrics (Hellinger, Gower, Mahalanobis-based correlation-aware KNN); contributed Numba optimizations yielding ~4x–27x speedups, including ~20x for Doane histograms.
- Customer & revenue analytics: Raised telecom customer-to-call linkage ~2%, segmented caller behavior, and built revenue-protection analytics across ~2M utility accounts.
- Enterprise decision support: Delivered reproducible pipelines and self-service analytics over ~11M corporate/partnership returns; expanded TY2024 analytical assets covering ~8.5M records.
- Automation & scalable delivery: Automated recurring forecast reporting and added checkpoint-and-restart to long-running geocoding.
- Governance & enablement: Validated R/Python solution parity, reviewed KPI logic, flagged source-data limitations, and shared modern analytics/AI methods across 3 communities of practice reaching 50–150 analysts weekly.

IITM Startup Embedded Wireless Software Engineer **Dec 1997- Jul 2001** Chennai, India

- Engineered DSP C and Assembly software for DECT wireless handsets, delivering capabilities in auto-ranging, over-the-air FLASH programming, and memory management

EDUCATION

University of Pittsburgh *Pittsburgh,PA*
Master of Science, Bioengineering

NIH imaging Fellow & Thesis: "Computational and experimental evaluation of the Tic-Tac-Toe RF coil for 7 Tesla MRI" study set out to solve two coupled problems that plague ultra-high-field (7T) neuroimaging.

George Washington University
Master of Science, Computer Engineering *Washington, DC*
GPA: 3.7 GPA**,Teaching Fellow** for Circuit Design, Signals and Systems, Digital and Analog Communication, and Stocha stic Processes.**Coursework:** VLSI, Embedded Electronics, Databases & JDBC

Regional Engineering College Warangal (now National Institute of Technology Warangal)
Bachelor of Technology, Mechanical Engineering *Andhra Pradesh, India*

CERTIFICATIONS

- Applications of Gen AI, John Hopkins University: Certificate (https://www.mygreatlearning.com/certificate/PUDGNRYB), Transcript (https://drive.google.com/file/d/1foN07jopU4TvXeMXuEefABZVCT0rjv7Q/view) (JAN 2026)
- Making Data-Driven Decisions, MIT IDSS, Schwarzman College: Certificate (https://www.mygreatlearning.com/certificate/KVYFRYEW), Transcript (-https://drive.google.com/file/d/1PmbdRrWsg67VPFLxojhweIvLX-qaN5Hq/view) (MAR 2025)
- Certificate in Computing: CDAC, India

SKILLS

- **Data Science & Modeling**: Predictive Modeling, Statistical Modeling, Machine Learning (ML), Natural Language Processing (NLP), Time-Series Analysis, Clustering, Random Forest, SVM, Multivariate Regression, Anomaly Detection, Topic Modeling, Change Point Analysis, Hypothesis Testing, Model Validation, Model Tuning, Feature Engineering, Tree-based Methods, Performance Metrics
- **Programming & Tools**: Python, SQL, R, Scikit-learn, PyTorch, Pandas, Teradata, Postgres SQL, Microsoft Azure, Databricks, Tableau, Power BI, MS SQL Server, C#/.NET, Docker, Git, ArcGIS ESRI tools
- **Data Engineering & Platforms**: Data Wrangling, Data Visualization, FedRAMP / Restricted Compute Environments, Cloud-based Systems, Synthetic Data Generation (SDV), Record Linkage, Semantic Search, Text Analytics